Independence

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1 Definitions

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

Definition 1 (Pairwise independence). *Events $A, B \in \mathcal{F}$ are* independent *iff*

$$P(A \cap B) = P(A)\,P(B).$$

Definition 2 (Mutual independence). *A finite family $\{A_1, \dots, A_n\} \subset \mathcal{F}$ is* mutually independent *iff for every non-empty subset $S \subseteq \{1, \dots, n\}$,*

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i).$$

Mutual independence is strictly stronger than pairwise independence.

Definition 3 (Conditional independence). *Given $C \in \mathcal{F}$ with $P(C) > 0$, events A, B are* conditionally independent given C *iff*

$$P(A \cap B \mid C) = P(A \mid C)\,P(B \mid C).$$

2 Main theorem

Theorem 1 (Independence of complements). *If A and B are independent, then so are A^c and B.*

Proof. Decompose B as a disjoint union $B = (A \cap B) \sqcup (A^c \cap B)$. By finite additivity,

$$P(B) = P(A \cap B) + P(A^c \cap B),$$

so

$$P(A^c \cap B) = P(B) - P(A \cap B).$$

Using independence of A and B,

$$P(A^c \cap B) = P(B) - P(A)\,P(B) = (1 - P(A))\,P(B) = P(A^c)\,P(B).$$

Hence A^c and B are independent. $\square$

Applying the theorem twice (first to (A, B), then to (B, A^c)) shows that A^c and B^c are also independent.

3 A pairwise-but-not-mutually-independent triple

Example 1. *Let* $\Omega = \{1, 2, 3, 4\}$ *with the uniform measure. Define*

$$A = \{1, 2\}, \quad B = \{1, 3\}, \quad C = \{1, 4\}.$$

Then $P(A) = P(B) = P(C) = \frac{1}{2}$*. Each pairwise intersection equals* $\{1\}$*, so* $P(A \cap B) = P(A \cap C) = P(B \cap C) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$*, giving pairwise independence. But* $A \cap B \cap C = \{1\}$*, so* $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8} = P(A)P(B)P(C)$*.*